Module 2

Applications of Partial Differential Equations

(Text 1 - Relevant portions of 18.3, 18.4, 18.5)

Syllabus :-

One dimensional wave equation – vibrations of a stretched string, derivation, solution of the wave equation using method of separation of variables, D'Alembert's solution of the wave equation, One dimensional heat equation, derivation, solution of the heat equation.

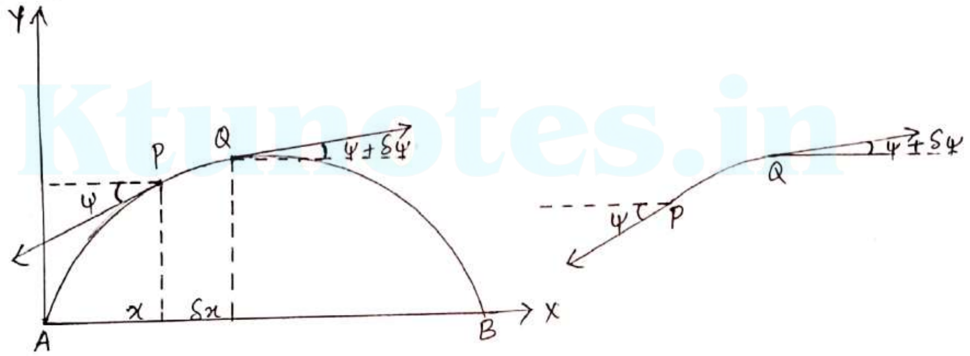
Vibrations of a stretched string – wave equation

We want to derive the PDE modeling small transverse vibrations of an elastic string, such as a violin string.

Consider a tightly stretched elastic string of length l and fixed ends A and B and subjected to constant tension T.
Take the end A as origin ($x=0$), AB as x-axis (at B, $x=l$) and AY perpendicular to it as y-axis; so that the motion takes place entirely in the xy-plane.

let the string be released from rest (time $t=0$) and allowed to vibrate. We shall study the subsequent motion of the string, with no external forces acting on it, assuming that each point of the string makes small vibrations at right angles to the equilibrium position AB, of the string entirely in one plane.

The problem is to determine the vibrations of the string, ie, to find its deflection $y(x,t)$ at any point x and at any time $t>0$.



$y(x,t)$ will be the solution of a PDE that is the model of the physical system to be derived. As the PDE should not be too complicated, so that we can solve it, we use reasonable simplifying <u>assumptions</u>:

1. The mass of the string per unit length is constant (homogeneous string). The string is perfectly elastic and does not offer any resistance to bending.

2. The tension caused by stretching the string before fastening it at the ends is so large that the action of the gravitational force on the string (trying to pull the string down a little) can be neglected.
3. The string performs small transverse motions in a vertical plane; ie, every particle of the string moves strictly vertically and so that the deflection (displacement) and the slope at every point of the string always remain small in absolute value.

Consider the motion of the element PQ of the string between its points $P(x, y)$ and $Q(x+\delta x, y+\delta y)$, where the tangents make angles ψ and $\psi + \delta\psi$ with the x-axis. Clearly, the element is moving upwards with the acceleration $\frac{\partial^2 y}{\partial t^2}$ and there is no motion in the horizontal direction.

In the vertical direction, we have two forces, namely, the vertical components $-T\sin\psi$ and $T\sin(\psi+\delta\psi)$ of T; here the minus sign appears because the component at P is directed downward.

Hence, the vertical component of the force acting on PQ $= T\sin(\psi+\delta\psi) - T\sin\psi$

$$= T\left[\sin(\psi+\delta\psi) - \sin\psi\right]$$

$$= T\left[\tan(\psi+\delta\psi) - \tan\psi\right], \text{ since } \psi \text{ is small}$$

$\left[\text{if } \theta \text{ is small, } \sin\theta \approx \frac{\sin\theta}{\cos\theta} \text{ since } \cos\theta \to 1\right]$

$$= T\left[\left\{\frac{\partial y}{\partial x}\right\}_{x+\delta x} - \left\{\frac{\partial y}{\partial x}\right\}_{x}\right]$$ since $\tan\psi$ and $\tan(\psi+\delta\psi)$ are the slopes of the string at x and $x+\delta x$.

If m be the mass per unit length of the string. Then by Newton's second law of motion, the resultant force is equal to the mass $m\delta x$ of the portion times the acceleration $\frac{\partial^2 y}{\partial t^2}$, evaluated at some point between x and $x+\delta x$; here 'm' is the mass of the undeflected string per unit length and δx is the length of the portion of the undeflected string.

ie, $$m\,\delta x \cdot \frac{\partial^2 y}{\partial t^2} = T\left[\left\{\frac{\partial y}{\partial x}\right\}_{x+\delta x} - \left\{\frac{\partial y}{\partial x}\right\}_{x}\right]$$

ie, $$\frac{\partial^2 y}{\partial t^2} = \frac{T}{m}\left[\frac{\left\{\frac{\partial y}{\partial x}\right\}_{x+\delta x} - \left\{\frac{\partial y}{\partial x}\right\}_{x}}{\delta x}\right]$$

Taking limits as $Q \to P$; ie, $\delta x \to 0$,

we get $$\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2} \quad \text{where } c^2 = \frac{T}{m} \qquad (1)$$

This is called the one-dimensional wave equation.

Note: One dimensional means that the eq. involves only one space variable x. Also the physical constant $\frac{T}{m}$ is denoted by c^2 (instead of c) to indicate that this constant is positive, a fact that will be essential to the form of the solutions.

Boundary conditions and initial conditions

Since the string is fastened at the ends $x=0$ and $x=l$, we have the two boundary conditions:

(a) $y(0,t)=0$, (b) $y(l,t)=0 \quad \forall\, t \geq 0$ —— (2)

Furthermore, the form of the motion of the string will depend on its initial deflection (deflection at $t=0$) and on its initial velocity. Thus the two initial conditions are:

(a) $y(x,0)=f(x)$ (b) $y_t(x,0)=g(x)$, $0 \leq x \leq l$ —— (3)

where $y_t = \frac{\partial y}{\partial t}$.

We now have to find a solution of the PDE (1) satisfying the conditions (2) and (3).

Solution of the wave equation by method of separation of variables

Assume that solution of $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$ —— (1) is of the form

$y(x,t) = X(x) \cdot T(t)$ where X is a function of x and T is a fn. of t only.

Then $\frac{\partial^2 y}{\partial t^2} = X \cdot T''$ and $\frac{\partial^2 y}{\partial x^2} = X'' \cdot T$

Substituting these in given equation (1),

$XT'' = c^2 X'' T$

ie, $\frac{X''}{X} = \frac{1}{c^2}\frac{T''}{T}$, the variables are now separated.

The left side depends only on x and the right side depends only on t. Hence both sides must be constant. k (say).

$$\therefore \quad \frac{X''}{X} = \frac{1}{c^2}\frac{T''}{T} = k.$$

Thus we get two ODEs,

$$X'' - kX = 0 \quad ; \quad T'' - kc^2 T = 0$$

$$\frac{d^2X}{dx^2} - kX = 0 \quad ; \quad \frac{d^2T}{dt^2} - kc^2 T = 0$$

Since k is arbitrary, the form of solutions changes for $k<0$, $k=0$, $k>0$.

Case 1 : $k<0$:- Suppose $k = -p^2$

Then $\frac{d^2X}{dx^2} + p^2 X = 0 \quad ; \quad \frac{d^2T}{dt^2} + p^2c^2 T = 0$

A.E $\quad m^2 + p^2 = 0 \qquad m^2 + p^2c^2 = 0$

$m = \pm ip \qquad m = \pm ipc$

Solu. is,

$$X = c_1 \cos px + c_2 \sin px \qquad T = c_3 \cos pct + c_4 \sin pct$$

Case 2: $k=0$

Then $\frac{d^2X}{dx^2} = 0 \qquad \frac{d^2T}{dt^2} = 0$

Solutions are

$$X = c_1 x + c_2 \qquad T = c_3 t + c_4$$

Case 3 : $k>0$:- Suppose $k = p^2$

Then $\frac{d^2X}{dx^2} - p^2 X = 0 \qquad \frac{d^2T}{dt^2} - p^2c^2 T = 0$

A.E. $\quad m^2 - p^2 = 0 \qquad m^2 - p^2c^2 = 0$

$m = \pm p \qquad m = \pm pc$

Solutions are

$$X = c_1 e^{px} + c_2 e^{-px} \qquad T = c_3 e^{pct} + c_4 e^{-pct}$$

Thus the possible solutions of ① are

$$y(x,t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos pct + c_4 \sin pct)$$

$$y(x,t) = (c_1 x + c_2)(c_3 t + c_4)$$

$$y(x,t) = (c_1 e^{px} + c_2 e^{-px})(c_3 e^{pct} + c_4 e^{-pct})$$

As we will be dealing with problems on vibrations, y must be a periodic function of x and t. Hence their solution must involve trigonometric terms. Thus the only possible suitable solution of wave equation is

$$y(x,t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos pct + c_4 \sin pct)$$

Problems

1. A tightly stretched string of length l is fixed at both ends. Find the displacement $u(x,t)$ if the string is given an initial displacement $f(x)$ and an initial velocity $g(x)$.

The equation of vibrating string is

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{——} \quad (1)$$

The boundary conditions are

$$y(0,t) = 0, \quad y(l,t) = 0 \quad \forall\, t \geq 0$$

Initial conditions are $y(x,0) = f(x)$, $y_t(x,0) = g(x)$.

Solution of (1) is of the form

$$y(x,t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos pct + c_4 \sin pct)$$

Applying the boundary conditions,

$$y(0,t) = 0 \implies 0 = c_1(c_3 \cos pct + c_4 \sin pct)$$

$$\implies c_1 = 0 \quad \left(\begin{array}{l}\text{if } c_3 \cos pct + c_4 \sin pct = 0 \\ \text{then } y(x,t) \equiv 0\end{array}\right)$$

$\therefore$ Solution becomes

$$y(x,t) = c_2 \sin px\,(c_3 \cos pct + c_4 \sin pct)$$

$$y(l,t) = 0 \implies 0 = c_2 \sin pl\,(c_3 \cos pct + c_4 \sin pct)$$

$$\implies \sin pl = 0 \quad (\because c_2 \neq 0, (c_3 \cos pct + c_4 \sin pct) \neq 0)$$

$$\implies pl = n\pi$$

$$\implies p = \frac{n\pi}{l}, \ (n \text{ integer}).$$

$$\therefore y(x,t) = \sin\left(\frac{n\pi x}{l}\right)\left(a_n \cos\left(\frac{n\pi ct}{l}\right) + b_n \sin\left(\frac{n\pi ct}{l}\right)\right), \ n = 1, 2, 3, \ldots$$

are solutions.

Then by principle of superposition of solutions.

$$y(x,t) = \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi ct}{l}\right) + b_n \sin\left(\frac{n\pi ct}{l}\right)\right] \sin\left(\frac{n\pi x}{l}\right) \quad \text{——} \quad (1)$$

{For negative integer n, we obtain essentially the same solutions, except for a minus sign, because $\sin(-\alpha) = -\sin\alpha$}

Applying initial conditions.

$y(x,0) = f(x) \Longrightarrow y(x,0) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{l}\right) = f(x)$

which is the half-range sine series for $f(x)$

$\therefore$ the coefficients a_n are given by.

$$a_n = \frac{2}{l}\int_0^l f(x)\sin\left(\frac{n\pi x}{l}\right)dx \quad \text{——— (2)}$$

Now $$\frac{\partial y}{\partial t} = \sum_{n=1}^{\infty} \frac{n\pi c}{l}\left[-a_n \sin\left(\frac{n\pi ct}{l}\right) + b_n\cos\left(\frac{n\pi ct}{l}\right)\right]\sin\left(\frac{n\pi x}{l}\right)$$

$y_t(x,0) = g(x) \Longrightarrow y_t(x,0) = \sum_{n=1}^{\infty} \frac{n\pi c}{l} b_n \sin\left(\frac{n\pi x}{l}\right) = g(x)$

which is the half range sine series for $g(x)$.

$\therefore$ the coefficients are given by

$$\frac{n\pi c}{l} b_n = \frac{2}{l}\int_0^l g(x)\sin\left(\frac{n\pi x}{l}\right)dx$$

$$\therefore b_n = \frac{2}{n\pi c}\int_0^l g(x)\sin\left(\frac{n\pi x}{l}\right)dx \quad \text{——— (3)}$$

$\therefore$ the required solution is (1) where a_n and b_n are given by (2) and (3)

2. A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form $y = a\sin\left(\frac{\pi x}{l}\right)$ from which it is released at time $t=0$. Show that the displacement of any point at a distance x from one end at time t is given by. $y(x,t) = a\sin\left(\frac{\pi x}{l}\right)\cos\left(\frac{\pi ct}{l}\right)$.

The vibration of the string is given by

$$\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2} \quad \text{——— (1)}$$

Boundary conditions are

$$y(0,t) = 0, \quad y(l,t) = 0 \qquad \forall\, t \geq 0$$

The initial conditions are

$y(x,0) = a\sin\left(\frac{\pi x}{l}\right)$, $y_t(x,0) = 0$

Now solution of ① is of the form

$$y(x,t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos pct + c_4 \sin pct)$$

$y(0,t) = 0 \Rightarrow c_1 = 0$

$\therefore y(x,t) = \sin px\,(A\cos pct + B\sin pct)$

$y(l,t) = 0 \Rightarrow \sin pl = 0 \Rightarrow p = \frac{n\pi}{l}$, (n, an integer)

$\therefore y(x,t) = \sin\left(\frac{n\pi x}{l}\right)\left[A\cos\left(\frac{n\pi ct}{l}\right) + B\sin\left(\frac{n\pi ct}{l}\right)\right]$, $n = 1, 2, 3, \ldots$

$\therefore y(x,t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right)\left[A_n\cos\left(\frac{n\pi ct}{l}\right) + B_n\sin\left(\frac{n\pi ct}{l}\right)\right]$

$$\frac{\partial y}{\partial t} = \sum_{n=1}^{\infty} \frac{n\pi c}{l}\left[-a_n \sin\left(\frac{n\pi ct}{l}\right) + b_n\cos\left(\frac{n\pi ct}{l}\right)\right]\sin\left(\frac{n\pi x}{l}\right)$$

$y_t(x,0) = 0 \Rightarrow \sum_{n=1}^{\infty} \frac{n\pi c}{l} b_n \sin\left(\frac{n\pi x}{l}\right) = 0$

$\Rightarrow b_n = 0 \quad \forall n$

$\therefore y(x,t) = \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi ct}{l}\right)\sin\left(\frac{n\pi x}{l}\right)$

$y(x,0) = a\sin\left(\frac{\pi x}{l}\right) \Rightarrow a\sin\left(\frac{\pi x}{l}\right) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{l}\right)$

$\Rightarrow a_1 = a$, $a_2 = a_3 = \cdots = 0$

$\therefore$ Required solution is

$$\underline{\underline{y(x,t) = a\cos\left(\frac{\pi ct}{l}\right)\sin\left(\frac{\pi x}{l}\right)}}$$

3. A tightly stretched string with fixed end points $x=0$ and $x=l$ is initially in a position given by $y = y_0 \sin^3\left(\frac{\pi x}{l}\right)$. If it is released from rest from this position, find the displacement $y(x,t)$.

The equation of the vibrating string is

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{——— ①}$$

The boundary conditions are $y(0,t) = 0$, $y(l,t) = 0$

The initial conditions are

$$y(x,0) = y_0 \sin^3\left(\frac{\pi x}{l}\right) \; ; \; y_t(x,0) = 0$$

Solution of ① is of the form

$$y(x,t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos pct + c_4 \sin pct)$$

$$y(0,t) = 0 \implies c_1 = 0$$

$$\therefore y(x,t) = \sin px\,(A \cos pct + B \sin pct)$$

$y(l,t) = 0 \implies p = \frac{n\pi}{l}$, n being an integer.

$$\therefore y(x,t) = \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi ct}{l}\right) + b_n \sin\left(\frac{n\pi ct}{l}\right)\right] \sin\left(\frac{n\pi x}{l}\right)$$

$$\frac{\partial y}{\partial t} = \sum_{n=1}^{\infty} \frac{n\pi c}{l} \left[-a_n \sin\left(\frac{n\pi ct}{l}\right) + b_n \cos\left(\frac{n\pi ct}{l}\right)\right] \sin\left(\frac{n\pi x}{l}\right)$$

$$y_t(x,0) = 0 \implies 0 = \sum_{n=1}^{\infty} \frac{n\pi c}{l} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$\implies b_n = 0$$

$$\therefore y(x,t) = \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi ct}{l}\right) \sin\left(\frac{n\pi x}{l}\right)$$

Now $y(x,0) = y_0 \sin^3\left(\frac{\pi x}{l}\right)$

$$\boxed{\begin{aligned} &\sin 3\theta = 3\sin\theta - 4\sin^3\theta \\ &\therefore \sin^3 x = \tfrac{1}{4}\left[3\sin x - \sin 3x\right] \end{aligned}}$$

$$\implies \frac{y_0}{4}\left[3\sin\left(\frac{\pi x}{l}\right) - \sin\left(\frac{3\pi x}{l}\right)\right] = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{l}\right)$$

$$\implies a_1 = \frac{3y_0}{4}, \; a_3 = \frac{-y_0}{4}, \; a_2 = a_4 = a_5 = \cdots = 0$$

$\therefore$ Required solution is

$$y(x,t) = \frac{3y_0}{4} \cos\left(\frac{\pi ct}{l}\right) \sin\left(\frac{\pi x}{l}\right) - \frac{y_0}{4} \cos\left(\frac{3\pi ct}{l}\right) \sin\left(\frac{3\pi x}{l}\right)$$

(4) A tightly stretched string with fixed end points $x=0$ and $x=l$ is initially at rest in its equilibrium position. If each of its points is given a velocity $\lambda x(l-x)$, find the displacement of the string at any distance x from one end at any time t.

The equation of the vibrating string is

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}. \quad \text{——— (1)}$$

The boundary conditions are $y(0,t)=0$; $y(l,t)=0$.

The initial conditions are $y(x,0)=0$; (initial displacement)

and $y_t(x,0) = \lambda x(l-x)$

Solution of (1) is of the form

$$y(x,t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos pct + c_4 \sin pct)$$

$$y(0,t) = 0 \Longrightarrow c_1 = 0$$

$$\therefore y(x,t) = \sin px\,(A \cos pct + B \sin pct)$$

$$y(l,t) = 0 \Longrightarrow p = \frac{n\pi}{l} \text{ , n being an integer}$$

$$\therefore y(x,t) = \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi ct}{l}\right) + b_n \sin\left(\frac{n\pi ct}{l}\right)\right] \sin\left(\frac{n\pi x}{l}\right)$$

$$y(x,0) = 0 \Longrightarrow 0 = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{l}\right) \Longrightarrow a_n = 0$$

$$\therefore y(x,t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi ct}{l}\right) \sin\left(\frac{n\pi x}{l}\right)$$

$$\frac{\partial y}{\partial t} = \sum_{n=1}^{\infty} \frac{n\pi c}{l} b_n \cos\left(\frac{n\pi ct}{l}\right) \sin\left(\frac{n\pi x}{l}\right)$$

$$y_t(x,0) = \lambda x(l-x) \Longrightarrow$$

$$\lambda x(l-x) = \sum_{n=1}^{\infty} \frac{n\pi c}{l} b_n \sin\left(\frac{n\pi x}{l}\right)$$ which is the half range sine series of $\lambda x(l-x)$ in $(0,l)$.

$\therefore$ the coefficients are given by

$$\frac{n\pi c}{l} b_n = \frac{2}{l}\int_0^l \lambda x(l-x)\sin\left(\frac{n\pi x}{l}\right)dx = \frac{2\lambda}{l}\int_0^l (lx-x^2)\sin\left(\frac{n\pi x}{l}\right)dx$$

$$= \frac{2\lambda}{l}\left[(lx-x^2)\left(\frac{-\cos\left(\frac{n\pi x}{l}\right)}{\frac{n\pi}{l}}\right) - (l-2x)\left(\frac{-\sin\left(\frac{n\pi x}{l}\right)}{\frac{n^2\pi^2}{l^2}}\right) + (-2)\cdot\left(\frac{\cos\left(\frac{n\pi x}{l}\right)}{\frac{n^3\pi^3}{l^3}}\right)\right]_0^l$$

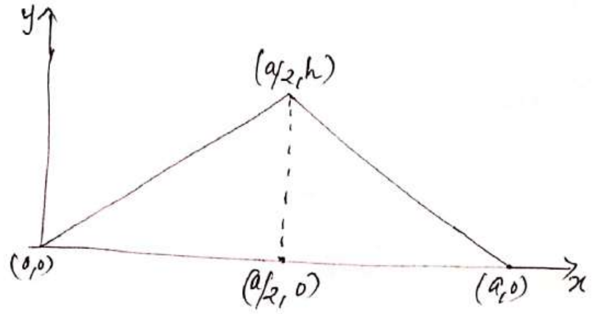
$$= \frac{2\lambda}{l}\left[\frac{-2l^3}{n^3\pi^3}\cos\left(\frac{n\pi x}{l}\right)\right]_0^l = \frac{4\lambda l^2}{n^3\pi^3}\left[1-(-1)^n\right]$$

$$\therefore b_n = \frac{4\lambda l^3}{cn^4\pi^4}\left(1-(-1)^n\right)$$

$\therefore$ the desired solution is

$$y(x,t) = \frac{4\lambda l^3}{c\pi^4}\sum_{n=1}^{\infty}\frac{\left[1-(-1)^n\right]}{n^4}\sin\left(\frac{n\pi ct}{l}\right)\sin\left(\frac{n\pi x}{l}\right)$$

5. A tightly stretched violin string of length a and fixed at both ends is plucked at its mid-point and assumes initially the shape of a triangle of height 'h'. Find the displacement $y(x,t)$ at any distance x and any time 't' after the string is released from rest.



The displacement of the string is given by

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{——— (1)}$$

Boundary conditions are $y(0,t)=0$, $y(a,t)=0$

Initial conditions are $y(x,0) = \begin{cases} \frac{2hx}{a}, & 0<x<\frac{a}{2} \\ \frac{2h}{a}(a-x), & \frac{a}{2}<x<a \end{cases}$

and $y_t(x,0)=0$.

Solution of (1) is of the form

$$y(x,t) = (c_1\cos px + c_2 \sin px)(c_3\cos pct + c_4 \sin pct)$$

$$y(0,t)=0 \Longrightarrow c_1 = 0$$

$$\therefore y(x,t) = \sin px\,(A\cos pct + B\sin pct)$$

$y(a,t)=0 \Longrightarrow p = \frac{n\pi}{a}$, n being an integer.

$$\therefore y(x,t) = \sum_{n=1}^{\infty}\left[a_n\cos\left(\frac{n\pi ct}{a}\right) + b_n \sin\left(\frac{n\pi ct}{a}\right)\right]\sin\left(\frac{n\pi x}{a}\right)$$

$$\frac{\partial y}{\partial t} = \sum_{n=1}^{\infty}\frac{n\pi c}{a}\left[-a_n\sin\left(\frac{n\pi ct}{a}\right) + b_n\cos\left(\frac{n\pi ct}{a}\right)\right]\sin\left(\frac{n\pi x}{a}\right)$$

$$y_t(x,0)=0 \Longrightarrow b_n = 0$$

$$\therefore y(x,t) = \sum_{n=1}^{\infty} a_n\cos\left(\frac{n\pi ct}{a}\right)\sin\left(\frac{n\pi x}{a}\right)$$

$$y(x,0) = \begin{cases} \frac{2hx}{a}, & 0<x<\frac{a}{2} \\ \frac{2h}{a}(a-x), & \frac{a}{2}<x<a \end{cases} = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{a}\right)$$ which is the half range sine series of $y(x,0)$.

where $a_n = \frac{2}{a}\int_0^a u(x,0)\sin\left(\frac{n\pi x}{a}\right)dx$

$$= \frac{2}{a}\int_0^{a/2}\frac{2hx}{a}\sin\left(\frac{n\pi x}{a}\right)dx + \frac{2}{a}\int_{a/2}^{a}\frac{2h}{a}(a-x)\sin\left(\frac{n\pi x}{a}\right)dx$$

$$= \frac{4h}{a^2}\left[x.\frac{-\cos\left(\frac{n\pi x}{a}\right)}{\frac{n\pi}{a}} - 1.\frac{-\sin\left(\frac{n\pi x}{a}\right)}{\left(\frac{n\pi}{a}\right)^2}\right]_0^{a/2} +$$

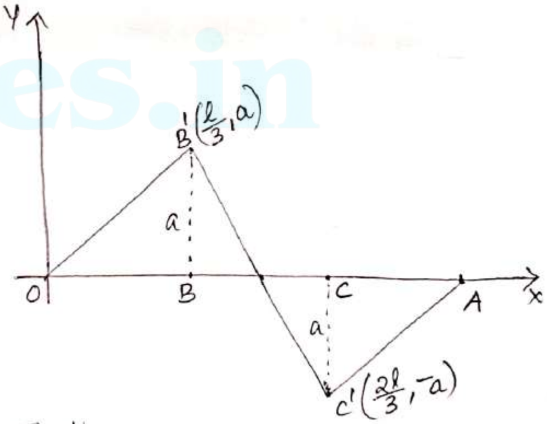
$$\frac{4h}{a^2}\left[(a-x).\frac{-\cos\left(\frac{n\pi x}{a}\right)}{\frac{n\pi}{a}} - -1.\frac{-\sin\left(\frac{n\pi x}{a}\right)}{\left(\frac{n\pi}{a}\right)^2}\right]_{a/2}^{a}$$

$$= \frac{4h}{a^2}\left[\cancel{\frac{-a}{n\pi}\cdot\frac{a}{2}\cos\left(\frac{n\pi}{2}\right)} + \frac{a^2}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right) + \cancel{\frac{a}{n\pi}\cdot\frac{a}{2}\cdot\cos\left(\frac{n\pi}{2}\right)} + \frac{a^2}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)\right]$$

$$= \frac{8h}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)$$

$$\therefore y(x,t) = \sum_{n=1}^{\infty}\frac{8h}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right).\cos\left(\frac{n\pi ct}{a}\right).\sin\left(\frac{n\pi x}{a}\right)$$

6. The points of trisection of a string are pulled aside through the same distance on opposite sides of ~~a~~ the position of equilibrium and the string is released from rest. Derive an expression for the displacement of the string at subsequent time and show that the mid-point of the string always remains at rest.



Let B and C be the points of trisection of the string OA $(=l)$.
Initially the string is held in the form $OB'C'A$, where $BB' = CC' = a$ (say).

The displacement $y(x,t)$ of the string is given by $\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2}$

Boundary conditions are $y(0,t)=0$; $y(l,t)=0$

Initial conditions are $y_t(x,0)=0$ and

$$y(x,0) = \begin{cases} \frac{3a}{l}x, & 0 \le x \le \frac{l}{3} \\ \frac{3a}{l}(l-2x), & \frac{l}{3} \le x \le \frac{2l}{3} \\ \frac{3a}{l}(x-l), & \frac{2l}{3} \le x \le l \end{cases}$$

The solution is of the form.

$$y(x,t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos pct + c_4 \sin pct)$$

Applying boundary conditions, we get $c_1 = 0$, $p = \frac{n\pi}{l}$

$$\therefore y(x,t) = \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi ct}{l}\right) + b_n \sin\left(\frac{n\pi ct}{l}\right)\right] \sin\left(\frac{n\pi x}{l}\right)$$

$$y_t(x,0) = 0 \Longrightarrow b_n = 0 \Longrightarrow y(x,t) = \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi ct}{l}\right) \sin\left(\frac{n\pi x}{l}\right)$$

Now $y(x,0) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{l}\right)$ where

$$a_n = \frac{2}{l}\int_0^l y(x,0) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{l}\left[\int_0^{l/3} \frac{3ax}{l} \cdot \sin\left(\frac{n\pi x}{l}\right) dx + \int_{l/3}^{2l/3} \frac{3a}{l}(l-2x)\sin\left(\frac{n\pi x}{l}\right) dx + \int_{2l/3}^{l} \frac{3a}{l}(x-l)\sin\left(\frac{n\pi x}{l}\right) dx\right]$$

$$= \frac{6a}{l^2}\left\{\left[x \cdot \frac{-\cos\left(\frac{n\pi x}{l}\right)}{\frac{n\pi}{l}} - 1 \cdot \frac{-\sin\left(\frac{n\pi x}{l}\right)}{\frac{n^2\pi^2}{l^2}}\right]_0^{l/3} + \left[(l-2x) \cdot \frac{-\cos\left(\frac{n\pi x}{l}\right)}{\frac{n\pi}{l}} - 2 \cdot \frac{-\sin\left(\frac{n\pi x}{l}\right)}{\frac{n^2\pi^2}{l^2}}\right]_{l/3}^{2l/3}\right.$$

$$\left. + \left[(x-l) \cdot \frac{-\cos\left(\frac{n\pi x}{l}\right)}{\frac{n\pi}{l}} - 1 \cdot \frac{-\sin\left(\frac{n\pi x}{l}\right)}{\frac{n^2\pi^2}{l^2}}\right]_{2l/3}^{l}\right\}$$

$$= \frac{6a}{l^2}\left\{\left[\cancel{\frac{-l^2}{3n\pi} \cos\left(\frac{n\pi}{3}\right)} + \frac{l^2}{n^2\pi^2}\sin\left(\frac{n\pi}{3}\right)\right] + \cancel{\frac{l^2}{3n\pi}\cos\left(\frac{2n\pi}{3}\right)} - \frac{2l^2}{n^2\pi^2}\sin\left(\frac{2n\pi}{3}\right)\right.$$

$$\left. + \cancel{\frac{l^2}{3n\pi}\cos\left(\frac{n\pi}{3}\right)} + \frac{2l^2}{n^2\pi^2}\sin\left(\frac{n\pi}{3}\right) + \left[\cancel{\frac{-l^2}{3n\pi}\cos\left(\frac{2n\pi}{3}\right)} - \frac{l^2}{n^2\pi^2}\sin\left(\frac{2n\pi}{3}\right)\right]\right\}$$

$$= \frac{6a}{\cancel{l^2}} \cdot \frac{3\cancel{l^2}}{n^2\pi^2}\left[\sin\left(\frac{n\pi}{3}\right) - \sin\left(\frac{2n\pi}{3}\right)\right] = \frac{18a}{n^2\pi^2}\left[\sin\left(\frac{n\pi}{3}\right) - \sin\left(n\pi - \frac{n\pi}{3}\right)\right]$$

$$= \frac{18a}{n^2\pi^2}\sin\left(\frac{n\pi}{3}\right)\left(1+(-1)^n\right)$$

$$\left(\because \sin 2n\pi = \sin\left(n\pi - \frac{n\pi}{3}\right) = -(-1)^n \sin\frac{n\pi}{3}\right)$$

$$= \begin{cases} 0, & n \text{ odd} \\ \frac{36a}{n^2\pi^2}\sin\left(\frac{n\pi}{3}\right), & n \text{ even} \end{cases}$$

$\therefore \; y(x,t) = \dfrac{36a}{\pi^2} \displaystyle\sum_{n=\text{even}}^{\infty} \dfrac{\sin\left(\frac{n\pi}{3}\right)}{n^2} \sin\left(\frac{n\pi x}{l}\right) \cos\left(\frac{n\pi c t}{l}\right)$

put $n = 2m$

Then $y(x,t) = \dfrac{9a}{\pi^2} \displaystyle\sum_{m=1}^{\infty} \dfrac{1}{m^2} \sin\left(\frac{2m\pi}{3}\right) \sin\left(\frac{2m\pi x}{l}\right) \cos\left(\frac{2m\pi c t}{l}\right)$

At the mid-point, $x = \frac{l}{2}$

then $\sin \frac{2m\pi x}{l} = \sin m\pi = 0 \quad \forall m$

$\therefore \; y(l/2, t) = 0$

ie, mid-point of the string is always at rest.

H.W

1. Solve the wave equation $\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$ under the conditions $u=0$ when $x=0$ and $x=\pi$, $\frac{\partial u}{\partial t} = 0$ when $t=0$ and $u(x,0) = x$, $0 < x < \pi$.

2. A tightly stretched string with fixed end points at $x=0$ and $x=1$, is initially in a position given by

$$f(x) = \begin{cases} x, & 0 \le x \le \frac{1}{2} \\ 1-x, & \frac{1}{2} \le x \le 1 \end{cases}$$

If it is released from this position with velocity 'a', perpendicular to the x-axis, find the displacement at any point x of the string at any time $t > 0$.

D' Alembert's solution of wave equation

The solution of wave equation

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{——— (1)}$$

can be written in closed form (an expression without any summation or product) which is obtained by transforming (1) in a suitable way, by introducing new independent variables,

$$u = x + ct, \quad v = x - ct$$

then $\frac{\partial y}{\partial x} = \frac{\partial y}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial y}{\partial v} \cdot \frac{\partial v}{\partial x}$ $\qquad y \begin{smallmatrix} u \\ v \end{smallmatrix} x, t$

$$= \frac{\partial y}{\partial u} + \frac{\partial y}{\partial v}$$

$$\frac{\partial^2 y}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial y}{\partial u} + \frac{\partial y}{\partial v}\right) = \frac{\partial}{\partial u}\left(\frac{\partial y}{\partial u} + \frac{\partial y}{\partial v}\right) + \frac{\partial}{\partial v}\left(\frac{\partial y}{\partial u} + \frac{\partial y}{\partial v}\right)$$

$$= \frac{\partial^2 y}{\partial u^2} + 2\frac{\partial^2 y}{\partial u \partial v} + \frac{\partial^2 y}{\partial v^2}$$

Similarly, $\frac{\partial^2 y}{\partial t^2} = c^2\left(\frac{\partial^2 y}{\partial u^2} - 2\frac{\partial^2 y}{\partial u \partial v} + \frac{\partial^2 y}{\partial v^2}\right)$

Substituting these in (1), we get, $\frac{\partial^2 y}{\partial u \partial v} = 0$

Integrating w.r.to v, $\frac{\partial y}{\partial u} = h(u)$

Integrating w.r.to u, $y = \int h(u)\,du + \psi(v)$

$$= \phi(u) + \psi(v) \quad \left(\because \int h(u)du \text{ is a function of } u \text{ alone}\right)$$

ie, $\boxed{y(x,t) = \phi(x+ct) + \psi(x-ct)}$ ——— (2)

This is the general solution of (1) which is known as d'Alembert's solution.

Applying the <u>initial conditions</u>,

ie, suppose initially $y(x,0) = f(x)$ and $y_t(x,0) = 0$

Differentiating ② w.r.to t,

$$y_t = c\phi'(x+ct) - c\psi'(x-ct)$$

$y_t(x,0) = 0 \Longrightarrow 0 = c\phi'(x) - c\psi'(x) \Longrightarrow \phi'(x) = \psi'(x)$ ——— ③

$y(x,0) = f(x) \Longrightarrow \phi(x) + \psi(x) = f(x)$ ——— ④

Integrating ③ w.r.to x, we get

$$\phi(x) = \psi(x) + k$$

$\therefore$ ④ $\Longrightarrow$ $2\psi(x) + k = f(x)$

$\therefore \psi(x) = \frac{1}{2}[f(x) - k]$ and $\phi(x) = \frac{1}{2}[f(x) + k]$

$\therefore y(x,t) = \phi(x+ct) + \psi(x-ct)$

$= \frac{1}{2}[f(x+ct) + k] + \frac{1}{2}[f(x-ct) - k]$

ie, $\boxed{y(x,t) = \frac{1}{2}[f(x+ct) + f(x-ct)]}$ which is the required d'Alembert's solution of wave equation

Ktunotes.in

Problems

1. Find the deflection of a vibrating string of unit length having fixed ends with initial velocity zero and initial deflection $f(x) = k(\sin x - \sin 2x)$.

Given $y(x,0) = k(\sin x - \sin 2x)$ and

$y_t(x,0) = 0$

By d'Alembert's method, the solution is

$y(x,t) = \frac{1}{2}[f(x+ct) + f(x-ct)]$

$= \frac{1}{2}\left[k\left(\sin(x+ct) - \sin(2x-2ct)\right) + k\left(\sin(x-ct) - \sin(2x-2ct)\right)\right]$

$= \frac{k}{\cancel{2}}\left[\cancel{2}\sin x \cos ct - \cancel{2}\sin 2x \cos 2ct\right]$

Also $y(x,0) = k(\sin x \cancel{\cos ct} - \sin 2x) = f(x)$ and

$y_t(x,0) = k(-c\sin x \sin ct + 2c \sin 2x \sin 2ct)_{t=0} = 0$

ie, boundary conditions are satisfied //.

2. Using d' Alembert's method, find the deflection of a vibrating string of unit length having fixed ends, with initial velocity zero and initial deflection

(i) $f(x) = a(x - x^2)$

H.W (ii) $f(x) = a \sin^2 \pi x$

(i) $f(x) = a(x - x^2)$

$$y(x,t) = \frac{1}{2}\left[f(x+ct) + f(x-ct)\right]$$

$$= \frac{1}{2}\left[a(x+ct - (x+ct)^2) + a(x-ct - (x-ct)^2)\right]$$

$$= \frac{a}{2}\left[2x - (2x^2 + 2c^2t^2)\right]$$

$$= \underline{\underline{a(x - x^2 - c^2t^2)}}$$

Also $y(x,0) = a(x - x^2) = f(x)$ and

$$y_t(x,t) = -2ac^2t$$

$$\therefore y_t(x,0) = 0$$

hence boundary $\underline{\underline{\text{conditions}}}$ are satisfied.

One dimensional heat equation

Heat equation is a PDE describing diffusion (distribution) of materials (may be pollutants or temperature or energy in the form of waves) in a medium. Here we consider transfer of energy in the form of heat from a region of higher temperature to a region of lower temperature.

The energy transfer may be along a straight line or over a plane area or in a solid region. Accordingly we have one dimensional or two dimensional or three dimensional energy transfer PDE.

Here we consider the transfer of heat along a thin rod of length 'l' and the resulting PDE is called one dimensional heat equation.

Derivation

Consider the heat flow along a uniform rod of length 'l' which is kept along the x-axis with one end at $O(0,0)$ and other end at $P(l,0)$ and the direction of flow is along the positive x-axis.

Let $u(x,t)$ be the temperature at any point of the rod (bar) at the distance x from the origin at time t. For the derivation of the PDE, we make use of the fundamental postulates of heat conduction.

① Heat energy moves from a region of higher temperature to a region of lower temperature.

② The quantity of heat crossing through an area of cross-section A of the bar per second is directly proportional to the area and the temperature gradient $\frac{\partial u}{\partial x}$ normal to the area.

$$Q = -KA\frac{\partial u}{\partial x} \text{ per sec}$$

The proportionality constant K is called the thermal conductivity of the material and the negative sign is attached because as x increases, u decreases.

③ The quantity of heat required to produce a given temperature change Δu in a body is proportional to the mass of the body and to the temperature change.

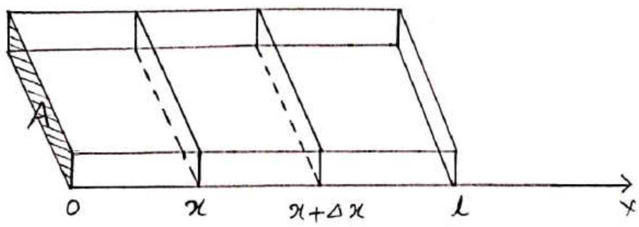
$$Q = S \, \text{Mass} \times \Delta u.$$

where the proportionality constant S is the specific heat capacity of the material.

The assumptions are:

① The rod is made up of homogeneous material with uniform cross section so that mass per unit length is constant and the temperature at any point in cross-section is same.

② The lateral faces are insulated so that heat is allowed to transfer along the length of the rod, not through the side faces.



Consider two cross sections of the rod at points P and Q which are x and $x+\Delta x$ distances from one end of the rod. Let Q_1 and Q_2 be the quantity of heat which flows through the cross sections at P and Q respectively.

The quantity of heat flows through the cross section at P in Δt time is

$$Q_1 = -KA\left(\frac{\partial u}{\partial x}\right)_x \Delta t$$

The quantity of heat flows through the cross section at Q in Δt time is

$$Q_2 = -KA\left(\frac{\partial u}{\partial x}\right)_{x+\Delta x} \Delta t$$

Hence the quantity of heat retained between the cross sections at P and Q is

$$Q_1 - Q_2 = KA\left[\left(\frac{\partial u}{\partial x}\right)_{x+\Delta x} - \left(\frac{\partial u}{\partial x}\right)_x\right]\Delta t$$

If the above quantity raises or lower the temperature of the rod by Δu, the quantity of heat increased or decreased is

$$S\,\text{Mass} \times \Delta u = \rho S A \Delta x \Delta u$$

where S is the specific heat capacity of the material and ρ, the density of the material.

$\therefore$ S Mass $\times \Delta u = \rho S A \Delta x \Delta u = KA\left[\left(\frac{\partial u}{\partial x}\right)_{x+\Delta x} - \left(\frac{\partial u}{\partial x}\right)_{x}\right]\Delta t$

$$\Longrightarrow \frac{\Delta u}{\Delta t} = \frac{K}{\rho S}\frac{\left[\left(\frac{\partial u}{\partial x}\right)_{x+\Delta x} - \left(\frac{\partial u}{\partial x}\right)_{x}\right]}{\Delta x}$$

Taking the limit $\Delta x \to 0$, we get

$$\frac{\partial u}{\partial t} = \frac{K}{\rho S}\frac{\partial^2 u}{\partial x^2}$$

ie, $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ —— ① where $c^2 = \frac{K}{\rho S}$ is known as diffusivity of the material of the bar.

This PDE is called the one dimensional heat transfer equation (diffusion equation)

Solution of heat equation

Using method of separation of variables, assume that the solution of ① is of the form

$$u(x,t) = X(x) \cdot T(t)$$

where X is a function of x alone and T is a function of t only.

Then $\frac{\partial u}{\partial x} = X'T$, $\frac{\partial^2 u}{\partial x^2} = X''T$, $\frac{\partial u}{\partial t} = XT'$

Then ① becomes $XT' = c^2 X'' T$

$$\Longrightarrow \frac{X''}{X} = \frac{T'}{c^2 T} = k \text{, say}$$

Then we get two ODEs.

$$X'' - kX = 0 \qquad \text{and} \qquad T' - kc^2 T = 0$$

The form of the solution changes for $k<0$, $k=0$ and $k>0$.

Case 1 :- $k > 0$

let $k = p^2$, Then

$X'' - p^2 X = 0$ and $T' - p^2c^2T = 0$

AE is $m^2 - p^2 = 0$ $\qquad$ $m - p^2c^2 = 0$

$m = \pm p$ $\qquad$ $m = p^2c^2$

$X = c_1 e^{px} + c_2 e^{-px}$ $\qquad$ $T = c_3 e^{p^2c^2t}$

$\therefore u(x,t) = XT = (c_1 e^{px} + c_2 e^{-px}) c_3 e^{p^2c^2t}$

as $t \to \infty$, $e^{p^2ht} \to \infty$ which shows unbounded temperature.

Case 2 :- $k = 0$

Then $X'' = 0$ and $T' = 0$

$\therefore X = c_1 x + c_2$ $\qquad$ $T = c_3$

$\therefore u(x,t) = (c_1 x + c_2) c_3$ which is independent of time that is not possible physically.

Case 3 :- $k < 0$

let $k = -p^2$

Then $X'' + p^2 X = 0$ and $T' + p^2c^2 T = 0$

AE $m^2 + p^2 = 0$ $\qquad$ $m + p^2c^2 = 0$

$m = \pm ip$ $\qquad$ $m = -p^2c^2$

$\therefore X = c_1 \cos px + c_2 \sin px$ $\qquad$ $T = c_3 e^{-p^2c^2t}$

$\therefore u(x,t) = (c_1 \cos px + c_2 \sin px) c_3 e^{-p^2c^2t}$

Here as t increases, temperature $u(x,t)$ decreases which corresponds to the physical nature of the problem.

$\therefore \boxed{u(x,t) = (c_1 \cos px + c_2 \sin px) c_3 e^{-p^2c^2t}}$

is the only suitable general solution of heat equation.

Note:- For heat equation, there are two boundary conditions are only one initial condition (since the equation contains only first order partial derivative w.r.to t.

The boundary conditions are

$$u(0,t) = a(t) \text{ and } u(l,t) = b(t) \quad \forall t \geq 0$$

In general, $a(t)$ and $b(t)$ can be functions of time or constants. But here we consider problems with $a(t)$ and $b(t)$ as some fixed temperature.

The initial condition is

$u(x,0) = f(x)$, $0 < x < l$ which is the initial distribution of temperature.

Steady state solution of heat equation

Here we consider a system in which the heat distribution is governed by one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ with the end points of the rod kept at some fixed temperatures for sufficiently long time. Then boundary conditions become

$$u(0,t) = a, \quad u(l,t) = b \quad \forall t \geq 0$$

When both the ends are kept at some constant temperatures for sufficiently long time, the system will reach a state in which each point of the rod (each cross-section) will attain a definite temperature which will not change any further as time varies. That is, the temperature at each point will be independent of time. This state of the system is called <u>steady state</u> of the system. If the temperature distribution is depending on time, the system is said to be in <u>transient state</u>. In this situation, the temperature distribution will be a function of time and space. The system will be in transient state for finite time in the beginning stage of the system. The system may attain steady state after sufficiently long time has elapsed.

<u>At steady state</u> $\frac{\partial u}{\partial t} = 0$, so that $\frac{\partial^2 u}{\partial x^2} = 0$

Integrating twice w.r.to x, $\boxed{u(x,t) = u(x) = c_1 x + c_2}$

where c_1 and c_2 are arbitrary constants.

$u(0,t) = a \Longrightarrow c_2 = a$

$u(l,t) = b \Longrightarrow c_1 = \frac{b-a}{l}$

$\therefore$ The required steady state solution is

$$u(x) = \frac{b-a}{l}x + a$$

<u>Problem</u>

1. Find the steady state temperature distribution in a rod of length 30cm, if the ends of the rod are kept at 20°c and 80°c.

Heat equation is $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$

In steady state temperature is independent of time.

$\therefore \frac{\partial u}{\partial t} = 0 \Longrightarrow \frac{\partial^2 u}{\partial x^2} = 0 \Longrightarrow u = ax + b$

$u(0,t) = 20 \Longrightarrow \underline{20 = b}$

$u(30,t) = 80 \Longrightarrow 80 = 30a + 20 \Longrightarrow a = 2.$

$\therefore \underline{\underline{u(x) = 2x + 20}}$ is the required temperature distribution.

Fourier series solution of heat equation

We can have 5 different cases depending on the boundary conditions. They are.

1. Both ends are kept at zero temperature.
2. The left end is at zero temperature and right end is insulated.
3. The left end is insulated and right end is kept at zero temperature.
4. Both the ends are insulated.
5. Ends are neither at zero temperatures nor insulated

1 Both ends are kept at zero temperature

Here the boundary conditions are

$u(0,t)=0 \quad , u(l,t)=0 \quad \forall\ t \geq 0$

Initial temperature distribution is

$$u(x,0)=f(x) \ , \ 0 \leq x \leq l$$

Most suitable solution of one dimensional heat equation

$$u(x,t) = (C_1 \cos px + C_2 \sin px)\, c_3 e^{-p^2c^2t}$$

$$= (A \cos px + B \sin px) e^{-p^2c^2t}$$

$u(0,t)=0 \Longrightarrow 0 = A e^{-p^2c^2t} \Longrightarrow A=0$

$\therefore u(x,t) = B \sin px \, e^{-p^2c^2t}$

$u(l,t)=0 \Longrightarrow 0 = B \sin pl \, e^{-p^2c^2t} \Longrightarrow \sin pl = 0$

$\Longrightarrow pl = n\pi \Longrightarrow p = \frac{n\pi}{l}$, n being integer

The values of p for each n are called eigen values and the corresponding solutions are called eigen functions

Thus for each n, we have a solution

$$u_n(x,t) = B_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{n^2\pi^2c^2}{l^2}t}$$

Clearly, a single solution will not satisfy the initial condition. Since the heat equation is a linear PDE, using the principle of super position, the general solution is

$$u(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{c^2n^2\pi^2}{l^2}t}$$

$u(x,0) = f(x) \Longrightarrow f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{l}\right)$ which is the half range sine series for $f(x)$ in $(0,l)$

where $B_n = \frac{2}{l}\int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$ ——— ①

$\therefore$ the required solution is

$u(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{c^2n^2\pi^2}{l^2}t}$ where coefficients are given by ①.

Problems

1. Find the temperature distribution in a rod of length 2m whose end points are maintained at zero temperature and the initial temperature is $f(x) = 100(2x - x^2)$.

The temperature distribution is

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Boundary conditions are $u(0,t) = 0$; $u(2,t) = 0$

Initial condition is $u(x,0) = 100(2x - x^2)$.

Using separation of variables, solution is

$$u(x,t) = (c_1 \cos px + c_2 \sin px)\, c_3 e^{-p^2c^2t}$$

$u(0,t) = 0 \Longrightarrow c_1 = 0 \Longrightarrow u(x,t) = B \sin px\, e^{-p^2c^2t}$

$u(2,t) = 0 \Longrightarrow 2p = n\pi \Longrightarrow p = \frac{n\pi}{2}$, n being integer.

$$\therefore u(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{2}\right) e^{-\frac{n^2\pi^2c^2t}{4}}$$

$$u(x,0) = 100\left(2x - x^2\right) \Longrightarrow 100\left(2x - x^2\right) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right)$$

where $b_n = \frac{2}{2}\int_0^2 100\left(2x - x^2\right)\sin\left(\frac{n\pi x}{2}\right)dx$

$$= 100\left[\underbrace{\left(2x - x^2\right)}_{\to 0}\cdot\frac{-\cos\left(\frac{n\pi x}{2}\right)}{\frac{n\pi}{2}} - \underbrace{(2-2x)}_{\to 0}\cdot\frac{-\sin\left(\frac{n\pi x}{2}\right)}{\frac{n^2\pi^2}{4}} + -2\cdot\frac{\cos\left(\frac{n\pi x}{2}\right)}{\frac{n^3\pi^3}{8}}\right]_0^2$$

$$= 100\times -2\times\frac{8}{n^3\pi^3}\left[\cos n\pi - \cos 0\right] = \frac{1600}{\pi^3 n^3}\left(1-(-1)^n\right)$$

$$\therefore u(x,t) = \frac{1600}{\pi^3}\sum_{n=1}^{\infty}\frac{\left(1-(-1)^n\right)}{n^3}\sin\left(\frac{n\pi x}{2}\right)e^{-\frac{n^2\pi c^2}{4}t}$$

2. A rod of length 30 cm has its ends A and B kept at 20°c and 80°c respectively until steady state temperature prevail. The temperature at each end is then suddenly reduced to zero temperature and kept so. Find the resulting temperature function $u(x,t)$ taking $x=0$ at A.

The temperature along the rod satisfies the equation $\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2}$ ——— ①

In steady state temperature distribution is independent of time.

$$\therefore \frac{\partial u}{\partial t} = 0$$

$$\therefore ① \Longrightarrow 0 = \frac{d^2u}{dx^2} \Longrightarrow u = ax + b$$

$$u(0,t) = 20 \Longrightarrow b = 20 \Longrightarrow u = ax + 20$$

$$u(30,t) = 80 \Longrightarrow 80 = 30a + 20 \Longrightarrow a = 2$$

$$\therefore u(x) = 2x + 20$$

$\therefore$ The initial condition is

$$u(x,0) = f(x) = 2x + 20, \quad 0 \le x \le 30$$

Also the boundary conditions are

$u(0,t)=0 \;;\; u(30,t)=0$

$\therefore$ the solution of heat equation is

$u(x,t) = (c_1\cos px + c_2\sin px)\, c_3 e^{-p^2c^2t}$

$u(0,t)=0 \Rightarrow c_1 = 0$

$u(30,t)=0 \Rightarrow p = \frac{n\pi}{30}$, n being integer.

$$\therefore u(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{30}\right) e^{-\frac{n^2\pi^2c^2}{900}t}$$

$$u(x,0) = 2x+20 \Longrightarrow 2x+20 = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{30}\right)$$

where $B_n = \frac{2}{30}\int_0^{30}(2x+20)\sin\left(\frac{n\pi x}{30}\right)dx$

$$= \frac{1}{15}\left[(2x+20)\cdot\frac{-\cos\left(\frac{n\pi x}{30}\right)}{\frac{n\pi}{30}} - (2)\cdot\frac{-\sin\left(\frac{n\pi x}{30}\right)}{\frac{n^2\pi^2}{900}}\right]_0^{30}$$

$$= \frac{1}{15}\cdot\frac{-30}{n\pi}\left[(2\times30+20)\cdot\cos(n\pi) - (20)\cos 0\right]$$

$$= \frac{40}{n\pi}\left(1-4(-1)^n\right)$$

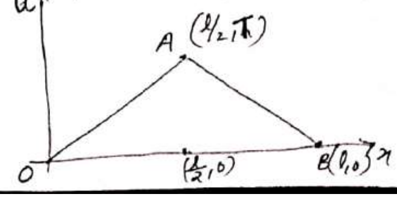
$$\therefore u(x,t) = \frac{40}{\pi}\sum_{n=1}^{\infty}\frac{[1-4(-1)^n]}{n}\sin\left(\frac{n\pi x}{30}\right)e^{-\frac{n^2\pi^2c^2}{900}t}$$

3. A homogeneous rod of conducting material of length l has its ends kept at zero temperature. The temperature at the centre is T and falls uniformly to zero at the two ends. Find the temperature $u(x,t)$.

$u(x,t)$ satisfies the equation $\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2}$

The boundary conditions are $u(0,t)=0 \;;\; u(l,t)=0$

Since the temperature at the centre is T and falls uniformly to zero at the two ends, its distribution at $t=0$ can be computed by finding the equations of OA and AB where $A\left(\frac{l}{2}, T\right)$ and $B(l,0)$



Equation of OA is $\quad u = \frac{2Tx}{l} \quad , 0 \le x \le \frac{l}{2}$

Equation of AB is $\quad u = \frac{2T}{l}(l-x), \ \frac{l}{2} \le x \le l$

$$\therefore u(x,0) = f(x) = \begin{cases} \frac{2Tx}{l} & , 0 \le x \le \frac{l}{2} \\ \frac{2T}{l}(l-x), & \frac{l}{2} \le x \le l \end{cases}$$

Solution of heat equation is

$$u(x,t) = (c_1 \cos px + c_2 \sin px)\, c_3 e^{-p^2c^2t}$$

$u(0,t) = 0 \implies c_1 = 0$

$u(l,t) = 0 \implies p = \frac{n\pi}{l}$, n being integer

$$\therefore u(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{c^2n^2\pi^2 t}{l^2}}$$

$u(x,0) = f(x) \implies f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{l}\right)$ where

$$B_n = \frac{2}{l}\int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{l}\left[\int_0^{l/2} \frac{2Tx}{l} \cdot \sin\left(\frac{n\pi x}{l}\right) dx + \int_{l/2}^{l} \frac{2T}{l}(l-x) \sin\left(\frac{n\pi x}{l}\right) dx\right]$$

$$= \frac{2}{l} \times \frac{2T}{l}\left[\left[x \cdot \frac{-\cos\left(\frac{n\pi x}{l}\right)}{\frac{n\pi}{l}} - 1 \cdot \frac{-\sin\left(\frac{n\pi x}{l}\right)}{\frac{n^2\pi^2}{l^2}}\right]_0^{l/2} + \left[(l-x)\cdot\frac{-\cos\left(\frac{n\pi x}{l}\right)}{\frac{n\pi}{l}} - (-1)\cdot\frac{-\sin\left(\frac{n\pi x}{l}\right)}{\frac{n^2\pi^2}{l^2}}\right]_{l/2}^{l}\right]$$

$$= \frac{4T}{l^2}\left[\cancel{-\frac{l}{2}\cdot\frac{l}{n\pi}\cos\left(\frac{n\pi}{2}\right)} + \frac{l^2}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right) + \cancel{\frac{l}{2}\frac{l}{n\pi}\cos\left(\frac{n\pi}{2}\right)} + \frac{l^2}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)\right]$$

$$= \frac{8T}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)$$

$$\therefore u(x,t) = \frac{8T}{\pi^2}\sum_{n=1}^{\infty}\frac{\sin\left(\frac{n\pi}{2}\right)}{n^2}\sin\left(\frac{n\pi x}{l}\right) e^{-\frac{c^2n^2\pi^2t}{l^2}}$$

4. Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with boundary conditions $u(x,0) = 3\sin(\pi x)$, $u(0,t) = 0$ and $u(1,t) = 0$ where $0 < x < 1$, $t > 0$.

The suitable solution of heat equation is

$$u(x,t) = (c_1 \cos px + c_2 \sin px)\, c_3 e^{-p^2 t}$$

$$= (A \cos px + B \sin px)\, e^{-p^2 t}$$

$u(0,t) = 0 \Rightarrow A = 0$

$$\Rightarrow u(x,t) = B \sin px\, e^{-p^2 t}$$

$u(1,t) = 0 \Rightarrow 0 = B \sin p\, e^{-p^2 t} \Rightarrow \sin p = 0 \Rightarrow p = n\pi$

Thus $u(x,t) = \sum_{n=1}^{\infty} B_n \sin(n\pi x)\, e^{-n^2 \pi^2 t}$

$u(x,0) = 3\sin(\pi x) \Rightarrow$

$$3\sin(\pi x) = \sum_{n=1}^{\infty} B_n \sin(n\pi x)$$

$$\Rightarrow B_1 = 3, \quad B_2 = B_3 = \dots = 0$$

$\therefore$ the required solution is

$$u(x,t) = 3\sin(\pi x)\, e^{-\pi^2 t}$$

5. Solve the one dimensional diffusion equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(x,t) \neq \infty$ if $t \to \infty$, $u(0,t) = 0 = u(\pi,t)$, $u(x,0) = \pi x - x^2$, $0 \leq x \leq \pi$

Heat equation is $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$

Boundary conditions are $u(0,t) = 0$ and $u(\pi,t) = 0$

Initial condition is $u(x,0) = \pi x - x^2$, $0 \leq x \leq \pi$

Since $u(x,t) \neq \infty$ if $t \to \infty$, the most suitable solution is $u(x,t) = (c_1 \cos px + c_2 \sin px)\, c_3 e^{-p^2 c^2 t}$

$$= (A \cos px + B \sin px)\, e^{-p^2 c^2 t}$$

$$u(0,t)=0 \Longrightarrow A=0$$

$$\Longrightarrow u(x,t) = B\sin px \, e^{-p^2c^2t}$$

$$u(\pi,t)=0 \Longrightarrow 0 = B\sin p\pi \, e^{-p^2c^2t} \Longrightarrow \sin p\pi = n\pi$$

$$\Longrightarrow p = n$$

$$\therefore u(x,t) = \sum_{n=1}^{\infty} B_n \sin nx \, e^{-n^2c^2t}$$

$$u(x,0) = \pi x - x^2, \quad 0 \le x \le \pi \Longrightarrow$$

$$\pi x - x^2 = \sum_{n=1}^{\infty} B_n \sin nx \quad \text{where}$$

$$B_n = \frac{2}{\pi}\int_0^{\pi} (\pi x - x^2)\sin nx \, dx$$

$$= \frac{2}{\pi}\left[\underbrace{(\pi x - x^2)}_{0} \cdot \frac{-\cos nx}{n} - \underbrace{(\pi - 2x)}_{0} \cdot \frac{-\sin nx}{n^2} + (-2)\cdot\frac{\cos nx}{n^3} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \times \frac{-2}{n^3}\left((-1)^n - 1\right) = \frac{4}{\pi n^3}\left(1 - (-1)^n\right)$$

$$\therefore u(x,t) = \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{\left(1-(-1)^n\right)}{n^3} \sin nx \, e^{-n^2c^2t}$$

(2) The left end is kept at 0°c and right end is insulated

One dimensional heat equation is

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}.$$

Assume that the end $x=0$ of the rod is kept at zero temperature and right end is insulated. By the law of heat conduction, the rate of heat flow is proportional to the gradient of temperature. Thus, if the end $x=l$ is insulated, then no heat can flow through that end. Hence the boundary conditions become

$$u(0,t)=0 \quad \text{and} \quad \frac{\partial u}{\partial x}(l,t)=0 \quad \forall\, t \geq 0$$

Initial condition is $u(x,0)=f(x)$, $0 \leq x \leq l$.

Now by method of separation of variables, suitable solution is

$$u(x,t) = (c_1 \cos px + c_2 \sin px)\, c_3$$

$$u(0,t)=0 \Longrightarrow c_1 = 0 \Longrightarrow u(x,t) = B \sin px\, e^{-p^2c^2t}$$

Then $\frac{\partial u}{\partial x} = Bp \cos px\, e^{-p^2c^2t}$.

$$\frac{\partial u}{\partial x}(l,t)=0 \Longrightarrow 0 = Bp\cos pl\, e^{-p^2c^2t}$$

$$\Longrightarrow \cos pl = 0 \Longrightarrow pl = (2n-1)\frac{\pi}{2}$$

$$\Longrightarrow p = (2n-1)\frac{\pi}{2l}, \text{ n being integer}$$

$$\therefore u(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{(2n-1)\pi x}{2l}\right) \cdot e^{-\frac{(2n-1)^2\pi^2c^2t}{4l^2}}$$

$u(x,0) = f(x) \Longrightarrow f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{(2n-1)\pi x}{2l}\right)$ which is the half range sine series for $f(x)$ and fourier coefficients are given by $B_n = \frac{2}{l}\int_0^l f(x) \sin\left(\frac{(2n-1)\pi x}{2l}\right) dx$.

Problem

1. Solve the boundary value problem $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ with the conditions $u(0,t) = 0 \;\forall\; t \geq 0$, $\frac{\partial u}{\partial x}(l,t) = 0$ and $u(x,0) = x$.

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Suitable solution is $u(x,t) = (c_1 \cos px + c_2 \sin px) c_3 e^{-p^2c^2t}$

$u(0,t) = 0 \Longrightarrow c_1 = 0 \Longrightarrow u(x,t) = B \sin px \, e^{-p^2c^2t}$

$\frac{\partial u}{\partial x} = Bp \cos px \, e^{-p^2c^2t}$

$\frac{\partial u}{\partial x}(l,t) = 0 \Longrightarrow 0 = Bp \cos pl \, e^{-p^2c^2t}$

$\Longrightarrow \cos pl = 0 \Longrightarrow p = \frac{(2n-1)\pi}{2l}$, n being integer

$$\therefore u(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{(2n-1)\pi x}{2l}\right) e^{-\frac{(2n-1)^2\pi^2c^2t}{4l^2}}$$

Now $u(x,0) = x \Longrightarrow x = \sum_{n=1}^{\infty} B_n \sin\left(\frac{(2n-1)\pi x}{2l}\right)$ where

$$B_n = \frac{2}{l}\int_0^l x \sin\left(\frac{(2n-1)\pi x}{2l}\right) dx$$

$$= \frac{2}{l}\left[x \cdot \frac{-\cos\left(\frac{(2n-1)\pi x}{2l}\right)}{\frac{(2n-1)\pi}{2l}} - 1 \cdot \frac{-\sin\left(\frac{(2n-1)\pi x}{2l}\right)}{\frac{(2n-1)^2\pi^2}{4l^2}} \right]_0^l$$

$$= \frac{2}{l} \cdot \frac{4l^2}{(2n-1)^2\pi^2} \sin\left(\frac{(2n-1)\pi}{2}\right) = \frac{8l}{\pi^2(2n-1)^2} \sin\left(\frac{(2n-1)\pi}{2}\right)$$

$$\therefore u(x,t) = \frac{8l}{\pi^2} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{(2n-1)\pi}{2}\right)}{(2n-1)^2} \sin\left(\frac{(2n-1)\pi x}{2l}\right) e^{-\frac{(2n-1)^2\pi^2c^2t}{4l^2}}$$

③ The left end is insulated and right end is kept at 0°C

Heat equation is $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$

Assume that the end $x=0$ of the rod is insulated and right end is kept at zero temperature. By law of heat conduction, the rate of heat flow is proportional to the gradient of the temperature normal to the area of cross-section. Thus, if the end $x=0$ is insulated, then no heat can flow through the end. Then boundary conditions become

$$\frac{\partial u}{\partial x}(0,t)=0 \quad \text{and} \quad u(l,t)=0 \quad \forall\, t \geq 0$$

Initial condition is $u(x,0)=f(x)$, $0 \leq x \leq l$

Solution of heat equation is

$$u(x,t) = (c_1 \cos px + c_2 \sin px)\, c_3 e^{-p^2c^2t}$$

Then $\frac{\partial u}{\partial x} = (-c_1 \sin px \cdot p + c_2 p \cos px)\, c_3 e^{-p^2c^2t}$

$$\frac{\partial u}{\partial x}(0,t)=0 \implies 0 = pB e^{-p^2c^2t} \implies B=0 \qquad B = c_2c_3.$$

$$\therefore u(x,t) = A \cos px \, e^{-p^2c^2t} \qquad A = c_1c_3.$$

$$u(l,t)=0 \implies 0 = A \cos pl \, e^{-p^2c^2t}$$

$$\implies \cos pl = 0 \implies p = \frac{(2n-1)\pi}{2l}, \text{ n being integer.}$$

$$\therefore u(x,t) = \sum_{n=1}^{\infty} A_n \cos\left(\frac{(2n-1)\pi x}{2l}\right) e^{-\frac{(2n-1)^2\pi^2c^2t}{4l^2}} \quad \text{———} \text{①}$$

$u(x,0)=f(x) \rightarrow f(x) = \sum_{n=1}^{\infty} A_n \cos\left(\frac{(2n-1)\pi x}{2l}\right)$ which is the half range cosine series for $f(x)$ in $(0,l)$

$$\therefore A_n = \frac{2}{l}\int_0^l f(x) \cos\left(\frac{(2n-1)\pi x}{2l}\right) dx \quad \text{———} \text{②}$$

$\therefore$ General solution is ① where A_n are given by ②.

Problem

1. Solve the boundary value problem $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ with the conditions $\frac{\partial u}{\partial x}(0,t)=0$, $u(l,t)=0$ $\forall\, t \geq 0$ and $u(x,0) = 20x$.

Heat equation is $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$

The solution is $u(x,t) = (c_1 \cos px + c_2 \sin px)\, c_3 e^{-p^2c^2t}$

$= (A\cos px + B \sin px)\, e^{-p^2c^2t}$

$\frac{\partial u}{\partial x} = (-Ap \sin px + Bp\cos px)\, e^{-p^2c^2t}$

$\frac{\partial u}{\partial x}(0,t) = 0 \Longrightarrow 0 = Bp \cdot e^{-p^2c^2t} \Longrightarrow B = 0$

$\therefore u(x,t) = A\cos px\, e^{-p^2c^2t}$

$u(l,t)=0 \Longrightarrow 0 = A\cos pl\, e^{-p^2c^2t} \Longrightarrow p = \frac{(2n-1)\pi}{2l}$, n being integer.

$\therefore u(x,t) = \sum_{n=1}^{\infty} A_n \cos\left(\frac{(2n-1)\pi x}{2l}\right) e^{-\frac{(2n-1)^2\pi^2c^2t}{4l^2}}$

$u(x,0) = 20x \Longrightarrow 20x = \sum_{n=1}^{\infty} A_n \cos\left(\frac{(2n-1)\pi x}{2l}\right)$

where $A_n = \frac{2}{l}\int_0^l 20x \cdot \cos\left(\frac{(2n-1)\pi x}{2l}\right) dx$

$$= \frac{2 \times 20}{l}\left[x \cdot \frac{\sin\left(\frac{(2n-1)\pi x}{2l}\right)}{\frac{(2n-1)\pi}{2l}} - 1 \cdot \frac{-\cos\left(\frac{(2n-1)\pi x}{2l}\right)}{\frac{(2n-1)^2\pi^2}{4l^2}} \right]_0^l$$

$$= \frac{40}{l}\left[l \cdot \frac{2l}{(2n-1)\pi} \sin\left(\frac{(2n-1)\pi}{2}\right) - \frac{4l^2}{(2n-1)^2\pi^2} \right]$$

$$= \frac{80l}{(2n-1)\pi}\left[\sin\left(\frac{(2n-1)\pi}{2}\right) - \frac{2}{(2n-1)\pi} \right]$$

$$\therefore u(x,t) = \frac{80l}{\pi} \sum_{n=1}^{\infty} \frac{\left[\sin\left(\frac{(2n-1)\pi}{2}\right) - \frac{2}{(2n-1)\pi}\right]}{(2n-1)} \cos\left(\frac{(2n-1)\pi x}{2l}\right) e^{-\frac{(2n-1)^2\pi^2c^2t}{4l^2}}$$

(4) When both ends are insulated

If the ends $x=0$ and $x=l$ are insulated, then no heat can flow through the end. Then boundary conditions are

$$\frac{\partial u}{\partial x}(0,t)=0 \quad \text{and} \quad \frac{\partial u}{\partial x}(l,t)=0 \quad \forall\, t\geq 0$$

Initial condition is $u(x,0)=f(x)$, $0\leq x\leq l$.

Solution of heat equation is

$$u(x,t)=(A\cos px+B\sin px)\,e^{-c^2p^2t}$$

$$\frac{\partial u}{\partial x}=(-Ap\sin px+Bp\cos px)\,e^{-p^2c^2t}$$

$$\frac{\partial u}{\partial x}(0,t)=0 \Longrightarrow 0=Bp\,e^{-p^2c^2t} \Longrightarrow B=0$$

$$\therefore u(x,t)=A\cos px\, e^{-c^2p^2t} \quad \text{——— (1)}$$

$$\frac{\partial u}{\partial x}=-Ap\sin px\cdot e^{-p^2c^2t}$$

$$\frac{\partial u}{\partial x}(l,t)=0 \Longrightarrow -Ap\sin pl\, e^{-p^2c^2t}=0$$

$$\Longrightarrow \sin pl=0 \Longrightarrow pl=n\pi \Longrightarrow p=\frac{n\pi}{l},\ n \text{ being integer}$$

From (1), we can see that corresponding to $n=0$, we get non-trivial solution and so for every non negative integer n, we get a solution.

$$u(x,t)=A_n\cos\left(\frac{n\pi x}{l}\right)e^{-\frac{n^2\pi^2c^2t}{l^2}}$$

$\therefore$ Most general solution is

$$u(x,t)=A_0+\sum_{n=1}^{\infty}A_n\cos\left(\frac{n\pi x}{l}\right)e^{-\frac{n^2\pi^2c^2t}{l^2}} \quad \text{——— (2)}$$

$u(x,0)=f(x) \Longrightarrow f(x)=A_0+\sum_{n=1}^{\infty}A_n\cos\left(\frac{n\pi x}{l}\right)$ which is the half range cosine series of $f(x)$ and hence the coefficients are given by

$$A_0=\frac{1}{l}\int_0^l f(x)\,dx \quad \text{—— (3)}; \qquad A_n=\frac{2}{l}\int_0^l f(x)\cos\left(\frac{n\pi x}{l}\right)dx \quad \text{——— (4)}$$

$\therefore$ General solution is (2) where A_0 and A_n are given by (3) and (4).

Problem

1 A. bar 100cm long, with insulated sides, has its ends kept at 0°c and 100°c until steady state conditions prevail. The two ends are then suddenly insulated and kept so. Find the temperature distribution in the rod after time t.

The temperature $u(x,t)$ on the rod satisfies the heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ —— ①

By law of heat conduction, the rate of heat flow is proportional to the gradient of the temperature. Thus if the ends $x=0$ and $x=l(=100)$ of the bar are insulated so that no heat can flow through the end. Then boundary conditions are

$$\frac{\partial u}{\partial x}(0,t) = 0 \quad ; \quad \frac{\partial u}{\partial x}(100,t) = 0 \quad \forall t \geq 0$$

Let $f(x)$ be the initial temperature in the bar. It can be found as follows:-

In steady state, the temperature distribution is independent of time.

$$\therefore \frac{\partial u}{\partial t} = 0$$

∴① becomes $\frac{d^2u}{dt^2} = 0 \Longrightarrow u = ax+b$

$u(0,t) = 0 \Longrightarrow b = 0 \Longrightarrow u = ax$

$u(100,t) = 100 \Longrightarrow 100 = a100 \Longrightarrow a = 1 \Longrightarrow u = x$

∴ The initial condition is

$$f(x) = u(x,0) = x \,, \quad 0 \leq x \leq 100$$

Now solution of ① is of the form

$$u(x,t) = (c_1 \cos px + c_2 \sin px)\, c_3 e^{-p^2c^2t}$$
$$= (A \cos px + B \sin px)\, e^{-p^2c^2t}$$

$$\frac{\partial u}{\partial x} = (-Ap\sin px + Bp\cos px)\, e^{-p^2c^2t}$$

$$\frac{\partial u}{\partial x}(0,t) = 0 \Longrightarrow 0 = Bp\, e^{-p^2c^2t} \Longrightarrow B = 0$$

$$\therefore u(x,t) = A\cos px\, e^{-p^2c^2t}$$

$$\frac{\partial u}{\partial x} = -Ap\sin px\, e^{-p^2c^2t}$$

$$\frac{\partial u}{\partial x}(100,t) = 0 \longrightarrow 0 = -Ap\sin 100p \cdot e^{-p^2c^2t}$$

$$\Longrightarrow \sin p100 = 0 \Longrightarrow 100p = n\pi$$

$$\Longrightarrow p = \frac{n\pi}{100}, \text{ n being integer}$$

Since, for $n=0$, we get a non trivial solution, the most general solution is

$$u(x,t) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{100}\right) e^{-\frac{n^2\pi^2c^2t}{100^2}}$$

Now $u(x,0) = x \Longrightarrow x = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{100}\right)$

Where the coefficients are given by

$$A_0 = \frac{1}{100}\int_0^{100} x\,dx = \frac{1}{100}\left[\frac{x^2}{2}\right]_0^{100} = \underline{\underline{50}}$$

$$A_n = \frac{2}{100}\int_0^{100} x\cos\left(\frac{n\pi x}{100}\right) dx$$

$$= \frac{1}{50}\left[x \cdot \frac{\sin\left(\frac{n\pi x}{100}\right)}{\frac{n\pi}{100}} - 1 \cdot \frac{-\cos\left(\frac{n\pi x}{100}\right)}{\frac{n^2\pi^2}{100^2}}\right]_0^{100}$$

$$= \frac{1}{50} \times \frac{100^2}{n^2\pi^2}\left((-1)^n - 1\right) = \frac{200}{n^2\pi^2}\left((-1)^n - 1\right)$$

$$\therefore u(x,t) = 50 + \frac{200}{\pi^2}\sum_{n=1}^{\infty} \frac{\left[(-1)^n - 1\right]}{n^2} \cos\left(\frac{n\pi x}{100}\right) e^{-\frac{n^2\pi^2c^2t}{10000}}$$

<u>Note</u>:- The solution of one dimensional heat equation depends on Fourier cosine series if either two ends are insulated or when left end is insulated and right end is kept at zero temperature.

(5) The ends are neither at zero temperatures nor insulated

If the end points of the rod are maintained at non-zero temperatures, the problem has non-zero boundary conditions (non-homogeneous boundary conditions) and so the system will have non-trivial steady state solution when sufficiently large time has elapsed. [But when the boundary conditions are zero or insulated (homogeneous boundary conditions), we can see that the steady state solution will become trivial solution. ie, $u(x,t) = u_s(x) = 0 \ \forall x$]
So in this case, the solution $u(x,t)$ will have two parts: a transient part $u_t(x,t)$ and a steady state part $u_s(x)$. So we take the solution in the form

$$u(x,t) = u_t(x,t) + u_s(x)$$

where $u_t(x,t)$ is the transient part solution whose boundary and initial conditions to be determined and $u_s(x)$ is the steady state solution with non-zero boundary conditions.

Clearly $u_t(x,t)$ satisfies the heat equation as it is the difference of two solutions of the heat equation.

Problems

1. A long iron rod with insulated lateral surface has its left end maintained at a temperature $0°c$ and its right end, at $x=2$, maintained at $100°c$. Determine the temperature as a function of x and t if the initial temperature is

$$u(x,0) = \begin{cases} 100x, & 0<x<1 \\ 100 & 1<x<2 \end{cases}$$

Here the endpoints of the rod are maintained at non-zero temperatures and hence the required temperature distribution in rod can be taken as

$$u(x,t) = u_s(x) + u_t(x,t) \quad \text{——— (1)}$$

where $u_s(x)$ is the steady state solution of heat equation involving x only and satisfying above non-zero boundary conditions and $u_t(x,t)$ is the transient part of the solution which decreases with increase of t.

The steady state temperature $u_s(x)$ is of the form $u_s = ax+b$ ($\because$ temperature is indepent of time $\frac{\partial u}{\partial t}=0 \Rightarrow \frac{\partial^2 u}{\partial x^2}=0$ $\Rightarrow u = ax+b$.)

The boundary conditions of u_s are $u_s(0)=0$ and $u_s(2)=100$.

$u_s(0)=0 \Longrightarrow 0=b \Longrightarrow u_s = ax$

$u_s(2)=100 \Longrightarrow 100=2a \Longrightarrow a=50$

$\therefore u_s(x) = 50x$ ——— (2)

Now from (1), $u_t(x,t) = u(x,t) - u_s(x)$

Putting $x=0$, $u_t(0,t) = u(0,t) - u_s(0)$

$= 0 - 0 = 0$

since the boundary conditions for $u(x,t)$ are $u(0,t)=0$ & $u(2,t)=100$

Putting $x=2$. $u_t(2,t) = u(2,t) - u_s(2,t)$

$$= 100 - 100$$
$$= 0$$

ie, $u_t(x,t)$ satisfies the homogeneous boundary conditions.

Also $u_t(x,0) = u(x,0) - u_s(x)$

$$= \begin{cases} 100x - 50x = 50x, & 0<x<1 \\ 100 - 50x, & 1<x<2 \end{cases} = f(x)$$

which is the initial condition for $u_t(x,t)$.

So we can find out $u_t(x,t)$ using the above boundary conditions and initial condition.

The suitable form of solution for u_t is given by

$$u_t(x,t) = (c_1 \cos px + c_2 \sin px)c_3 e^{-p^2c^2t}$$

Applying Boundary conditions,

$u_t(0,t)=0 \Rightarrow c_1 = 0$ and $u_t(2,t)=0 \Rightarrow p = \frac{n\pi}{2}$

$$\therefore u_t(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{2}\right) e^{-\frac{n^2\pi^2c^2t}{4}}$$

$u_t(x,0) = f(x) \Longrightarrow f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{2}\right)$ which is the half range sine series for $f(x)$ in $(0,2)$

$$\therefore B_n = \frac{2}{2}\int_0^2 f(x) \sin\left(\frac{n\pi x}{2}\right) dx$$

$$= \int_0^1 50x \cdot \sin\left(\frac{n\pi x}{2}\right) dx + \int_1^2 (100-50x) \sin\left(\frac{n\pi x}{2}\right) dx$$

$$= 50\left[x \cdot \frac{-\cos\left(\frac{n\pi x}{2}\right)}{\frac{n\pi}{2}} - 1 \cdot \frac{-\sin\left(\frac{n\pi x}{2}\right)}{\frac{n^2\pi^2}{4}}\right]_0^1 + 50\left[(2-x)\frac{-\cos\left(\frac{n\pi x}{2}\right)}{\frac{n\pi}{2}} - (-1)\frac{-\sin\left(\frac{n\pi x}{2}\right)}{\frac{n^2\pi^2}{4}}\right]_1^2$$

$$= 50\left[\cancel{\frac{-2}{n\pi}\cos\left(\frac{n\pi}{2}\right)} + \frac{4}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right) + \cancel{\frac{2}{n\pi}\cos\left(\frac{n\pi}{2}\right)} + \frac{4}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)\right]$$

$$= \frac{400}{\pi^2 n^2} \sin\left(\frac{n\pi}{2}\right)$$

$$\therefore u_t(x,t) = \sum_{n=1}^{\infty} \frac{400}{\pi^2 n^2} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi x}{2}\right) e^{-\frac{n^2\pi^2c^2t}{4}} \quad \text{--- (3)}$$

From (1), (2) and (3), the required temperature distribution is given by

$$u(x,t) = 50x + \frac{400}{\pi^2} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{n\pi}{2}\right)}{n^2} \sin\left(\frac{n\pi x}{2}\right) e^{-\frac{n^2\pi^2c^2t}{4}}$$

2. A bar 10cm long with insulated sides has its ends A and B maintained at 50°c and 100°c respectively until steady state conditions prevail. The temperature at A is suddenly raised to 90°c and at the same time that at B is lowered to 60°c. Find the temperature distribution in the bar at time t.

The temperature distribution $u(x,t)$ along the bar satisfies the heat equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Prior to the temperature change, when $t=0$, the bar is kept until steady state conditions prevail and so the heat flow was independent of time.

$$\therefore \frac{\partial u}{\partial t} = 0 \implies \frac{d^2u}{dx^2} = 0 \implies u = ax+b$$

$$u(0,t) = 50 \implies 50 = b \implies u = ax+50$$

$$u(10,t) = 100 \implies 100 = 10a+50 \implies a = 5$$

$$\therefore u = 5x+50, \quad 0 \le x \le 10$$

For the subsequent unsteady flow, the boundary conditions and initial condition are

$$u(0,t) = 90 \quad ; \quad u(10,t) = 60 \quad \forall\, t \ge 0$$

$$u(x,0) = 5x+50, \quad 0 \le x \le 10$$

Since the boundary conditions are non-zero, the required temperature distribution can be taken as

$$u(x,t) = u_s(x) + u_t(x,t) \quad \text{——— (1)}$$

where $u_s(x)$ is the steady state temperature distribution and $u_t(x,t)$ is the transient temperature distribution which decreases as t increases.

In steady state, $u_s(x) = ax + b$

$u_s(0) = 90^\circ \Longrightarrow 90 = b \Longrightarrow u_s(x) = ax + 90$

$u_s(10) = 60 \Longrightarrow 60 = 10a + 90 \Longrightarrow a = -3$

$$\therefore u_s(x) = -3x + 90 \quad \text{——— (2)}$$

Now consider $u_t(x,t) = u(x,t) - u_s(x)$

Putting $x = 0$, $u_t(0,t) = u(0,t) - u_s(0)$

$= 90 - 90 = 0$

Putting $x = 10$, $u_t(10,t) = u(10,t) - u_s(10)$

$= 60 - 60 = 0$

ie, $u_t(x,t)$ satisfies homogeneous boundary conditions.

Also $u_t(x,0) = u(x,0) - u_s(x)$

$= 5x + 50 - (-3x + 90) = 8x - 40$

which is the initial condition for u_t.

Now the suitable form of solution for u_t is

$$u_t(x,t) = (c_1 \cos px + c_2 \sin px)\, c_3\, e^{-p^2c^2t}$$

$u_t(0,t) = 0 \Rightarrow c_1 = 0$ and $u_t(10,t) = 0 \Rightarrow p = \frac{n\pi}{10}$

$$\therefore u_t(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{10}\right) e^{-\frac{n^2\pi^2c^2t}{100}}$$

$u_t(x,0) = 8x - 40 \Longrightarrow 8x - 40 = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{10}\right)$

where $B_n = \frac{2}{10}\int_0^{10}(8x-40)\sin\left(\frac{n\pi x}{10}\right)dx$

$$= \frac{8}{5}\left[(x-5)\cdot\frac{-\cos\left(\frac{n\pi x}{10}\right)}{\frac{n\pi}{10}} - 1\cdot\frac{-\sin\left(\frac{n\pi x}{10}\right)}{\frac{n^2\pi^2}{100}}\right]_0^{10}$$

$$= \frac{8}{5}\cdot\frac{-10}{n\pi}\left[5(-1)^n+5\right] = \frac{-80}{n\pi}\left[(-1)^n+1\right]$$

$$\therefore u_t(x,t) = \frac{-80}{\pi}\sum_{n=1}^{\infty}\frac{\left[(-1)^n+1\right]}{n}\sin\left(\frac{n\pi x}{10}\right)e^{-\frac{n^2\pi^2c^2t}{100}} \quad (3)$$

$\therefore$ from (1), (2) and (3),

$$u(x,t) = u_s(x) + u_t(x,t)$$

$$= 90-3x-\frac{80}{\pi}\sum_{n=1}^{\infty}\frac{\left((-1)^n+1\right)}{n}\sin\left(\frac{n\pi x}{10}\right)e^{-\frac{n^2\pi^2c^2t}{100}}$$

H.W

1. Find the temperature distribution in a bar of length π whose surface is thermally insulated with end points maintained at $0°c$. The initial temperature distribution is $u(x,0) = f(x) = \begin{cases} x, & 0 \le x \le \pi/2 \\ \pi - x, & \frac{\pi}{2} \le x \le \pi. \end{cases}$

2. Solve the PDE $\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2}$ for the conduction of heat along a road without radiation, subject to the following conditions: (1) u is not infinite for $t\to\infty$. (2) $\frac{\partial u}{\partial x} = 0$ for $x=0$ and $x=l$, (3) $u = lx - x^2$ for $t=0$ between $x=0$ and $x=l$.

3. An insulated rod of length l has its ends A and B maintained at $0°c$ and $100°c$ respectively until steady state conditions prevail.
   (a) If B is suddenly reduced to $0°c$ and maintained at $0°c$. find the temperature at a distance x from A at time t.
   (b) Solve the above problem if the change consists of raising the temperature of A to $20°c$ and reducing that of B to $80°c$.